Byeonghyun Pak

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Education

University of Maryland, College Park — Aug. 2026 – Present
Ph.D., Computer Science — *College Park, MD*
Advisor: *Prof. Furong Huang*

Daegu Gyeongbuk Institute of Science and Technology — Feb. 2019 – Feb. 2023
B.S., Computer Science — *Daegu, South Korea*
Advisor: *Prof. Kyong Hwan Jin*

Publications

Conference Papers

[1] Byeongju Woo, Zilin Wang, Byeonghyun Pak, Sangwoo Mo, and Stella X. Yu. **"Aligning Forest and Trees in Images & Long Captions for Visually Grounded Understanding"**. In *Conference on Neural Information Processing Systems* (**NeurIPS**), 2026. [paper]

[2] Sunghwan Kim*, Byeonghyun Pak*, Kehan Long, Yulun Tian, and Nikolay Atanasov. **"SERF: Spatiotemporal Environment and Robot Feature Map for Long-Horizon Mobile Manipulation"**. In *Conference on Robot Learning* (**CoRL**), 2026. **Spotlight**. [paper]

[3] Yunghee Lee, Byeonghyun Pak, Junwha Hong, and Hoseong Kim. **"Tortoise and Hare Guidance: Accelerating Diffusion Model Inference with Multirate Integration"**. In *Conference on Neural Information Processing Systems* (**NeurIPS**), 2025. [paper]

[4] Byeonghyun Pak*, Byeongju Woo*, Sunghwan Kim*, Dae-hwan Kim, and Hoseong Kim. **"Textual Query-Driven Mask Transformer for Domain Generalized Segmentation"**. In *European Conference on Computer Vision* (**ECCV**), 2024. [paper]

[5] Byeonghyun Pak*, Jaewon Lee*, and Kyong Hwan Jin. **"B-spline Texture Coefficients Estimator for Screen Content Image Super-Resolution"**. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2023. **Highlight Paper**. [paper]

Workshop Papers & Preprints

[6] Sunghwan Kim*, Byeonghyun Pak*, Kehan Long, Yulun Tian, and Nikolay Atanasov. **"4D Latent Mapping for Mobile Manipulation Policy Learning"**. In *ICRA 2026 Workshop on Multi-Modal Spatial AI*, 2026. **Best Paper Award**. [paper]

[7] Yunghee Lee and Byeonghyun Pak. **"Rethinking FID Through the Geometry of the Reference Dataset"**. In *ICML 2026 Workshop on Combining Theory and Benchmarks*, 2026. [paper]

[8] Seokmin Lee, Yunghee Lee, Byeonghyun Pak, and Byeongju Woo. **"Pixel-level Scene Understanding in One Token: Visual States Need What-is-Where Composition"**. In *CVPR 2026 Workshop on Pixel-level Video Understanding in the Wild*, 2026. [paper]

Research Experience

University of Maryland — Aug. 2026 – Present
Graduate Research Assistant — *College Park, MD*
Advisor: *Prof. Furong Huang*

- Developing self-improving agents for long-horizon mobile manipulation and embodied lifelong learning. Investigating agentic frameworks that accumulate world and skill knowledge, orchestrate reusable skills, and recover from failures across tasks and environments.

Independent Research — Jun. 2025 – Jun. 2026
Research Collaboration with UC San Diego — *Daejeon, South Korea*
Advisor: *Prof. Nikolay Atanasov*

- Conducted research on robot learning for long-horizon mobile manipulation. Developed methods combining vision-language-action models with 4D latent mapping to maintain persistent spatiotemporal context for policy learning. The resulting work was published at CoRL 2026 and received the Best Paper Award at an ICRA 2026 Workshop.

Agency for Defense Development Jun. 2023 – May 2026
Research Officer for National Defense *Daejeon, South Korea*
Advisor: *Dr. Hoseong Kim*

- Conducted research on robust visual learning and perception for autonomous systems. Led research on vision-language models for domain-generalized semantic segmentation, studied efficient diffusion models for synthetic data generation, and developed real-time perception systems for unmanned aerial vehicles.

Image Processing Lab, DGIST Dec. 2021 – Feb. 2023
Undergraduate Research Intern *Daegu, South Korea*
Advisor: *Prof. Kyong Hwan Jin*

- Conducted research on deep learning for image and signal processing, with a focus on implicit neural representations. Developed a B-spline-based neural representation for screen-content image super-resolution, resulting in a CVPR 2023 Highlight paper.

Honors & Awards

University of Maryland Dean's Fellowship 2026
- Awarded to outstanding incoming Ph.D. students

Best Paper Award, ICRA Workshop on Multi-Modal Spatial AI 2026
- Awarded for outstanding research presented at the workshop

Korea National Scholarship of Excellence in Science and Technology 2021 – 2023
- National full-tuition scholarship for outstanding undergraduate students

Korea National Scholarship for Undergraduate Study 2019 – 2023
- National full-tuition scholarship with living stipend

Korea Military Academy Superintendent's Award 2021
- Awarded for a research project on object-tracking models for defense systems

Academic Services

Journal Reviewer
- IEEE Transactions on Medical Imaging (**TMI**) 2025
- IEEE Robotics and Automation Letters (**RA-L**) 2026

Conference Reviewer
- Conference on Neural Information Processing Systems (**NeurIPS**) 2025, 2026
  - **Top Reviewer** (2026)
- Conference on Robot Learning (**CoRL**) 2026
  - **Outstanding Reviewer** (2026)
- Conference on Computer Vision and Pattern Recognition (**CVPR**) 2026
- Conference on Artificial Intelligence (**AAAI**) 2026
- Asian Conference on Computer Vision (**ACCV**) 2026

Workshop Reviewer
- **ICRA Workshop** on Semantics for Reliable Robot Autonomy 2026
- **IROS Workshop** on AI Meets Autonomy 2026

Skills

- **Programming Languages:** Python, C/C++, JavaScript, MATLAB
- **Frameworks & Tools:** PyTorch, TensorFlow, Docker, Git, OpenCV, OpenGL, Open3D
- **Languages:** Korean (native), English (fluent)